WEEK 4 · LESSON 19 OF 24 · HANDOUT

Bias and Fairness

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=UG_X_7g63rY — open the lesson online to watch inside the course.

🎓 **Go deeper:** Crash Course AI: Algorithmic Bias and Fairness — https://www.youtube.com/watch?v=gV0_raKR2UQ

Your goal in this lesson: Explain how bias enters AI systems, describe real cases of unfair AI, and apply practical steps to make a model fairer.

When a model is unfair

An AI system is **biased** (in the fairness sense) when it gives systematically worse results for some groups of people — by gender, age, region, language, skin colour, disability or income. A model does not need to be "told" to discriminate. It learns patterns from data, and if the data reflects an unfair past, the model can repeat and even amplify it.

Real cases

- **Face analysis:** the *Gender Shades* study (2018) by Joy Buolamwini and Timnit Gebru tested commercial gender-classification systems. Error rates were under 1% for lighter-skinned men but far higher — over 30% in one system — for darker-skinned women. The training images had included far fewer darker-skinned faces.
- **Hiring:** Amazon stopped using an experimental CV-screening tool after finding it rated CVs lower when they mentioned the word "women's" (as in "women's football club"). It had learned from ten years of past hiring, mostly of men.
- **Language:** many AI tools work well in English and poorly in Kinyarwanda, Kirundi or Luganda, so speakers of those languages get worse service.

Where bias comes from

Source	What happens	Example
Historical bias	The data reflects past unfairness	Past loans given mostly to men
Representation bias	Some groups are missing or rare in the data	Few rural farmers in a crop dataset
Measurement bias	A feature measures groups differently	Using "number of arrests" as a stand-in for "crime"
Labelling bias	The people labelling bring their own assumptions	Annotators judging "professional" photos
Deployment bias	A model is used in a setting it was not built for	A model trained in Kigali used unchanged in a rural district

Proxy features

Removing a sensitive column (such as gender) is **not enough**. Other features can act as **proxies** — they carry the same information indirectly. A first name, a school attended, or a home district can reveal gender, ethnicity or income.

Measuring fairness

Check results **separately for each group**, not only overall:

- Is the **accuracy** similar for women and men, urban and rural users?
- Is the **recall** similar? (Does the model catch high-risk patients equally in every district?)
- Is the **approval rate** fair, given real differences?

A model with 92% overall accuracy might be 96% accurate for one group and 75% for another. The average hides the harm.

Practical steps to reduce bias

1. **Ask who could be harmed** at the very start of the project.
2. **Collect representative data** — include the groups the model will serve.
3. **Measure performance per group** before deployment, and keep measuring after.
4. **Involve the affected community** and domain experts in design and testing.
5. **Keep a human in the loop** for high-stakes decisions such as loans, jobs, health and justice.
6. **Document** the data, its limits and the fairness checks (a "model card").

Remember this: Bias enters through data, labels, measurements and use. Removing sensitive columns is not enough. Measure results per group, use representative data, and keep humans responsible for serious decisions.

Name: _____ Date: _____

Activity

Task 1

Pick a model idea from your Week 2 plan.

Task 2

List three groups that might get worse results, and why.

Task 3

Name one proxy feature in your data that could reveal a sensitive attribute.

Task 4

Write the per-group checks you would run before deployment.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. When is an AI system biased in the fairness sense?
   - A. When it has many layers
   - B. When it uses colour
   - C. When it is slow
   - D. When it gives systematically worse results for some groups of people
2. What did the Gender Shades study find?
   - A. Face recognition never makes mistakes
   - B. Error rates were much higher for darker-skinned women than for lighter-skinned men
   - C. All systems were equally accurate
   - D. Men were always misclassified
3. Training data contains very few rural farmers. This is…
   - A. Overfitting only
   - B. Representation bias
   - C. Gradient descent
   - D. Data leakage
4. Why is deleting the gender column not enough to stop gender bias?
   - A. Gender columns cannot be deleted
   - B. Other features can act as proxies for gender
   - C. Models do not use columns
   - D. It is always enough
5. A model is 92% accurate overall. What should you also check?
   - A. Nothing more
   - B. Accuracy and recall for each group separately
   - C. The colour of the dashboard
   - D. Only the training speed
6. Which step helps reduce bias in high-stakes decisions?
   - A. Use less data
   - B. Hide the model's results
   - C. Remove all humans
   - D. Keep a human in the loop and measure results per group

© WAMUGII TECH SOLUTIONS · wamugii.kento.rw · Original course material — for enrolled students.